# Web Technology I

## Unit 6 - Introduction to Server Side and Client-Side Scripting

**Unit 6. Introduction to Server Side and Client-Side Scripting [2 Hrs.]**

**6.1. Overview of Server Side and Client-Side Scripting**

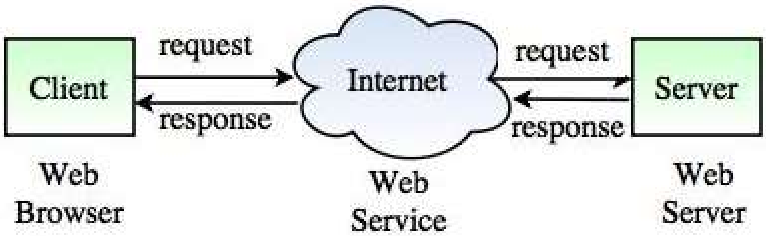
**6.2. Difference between Server Side and Client-Side Scripting**

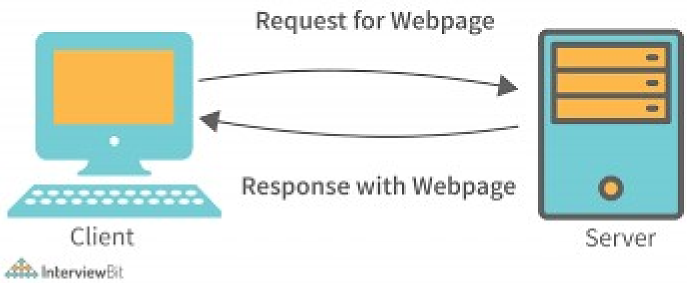
**6.3. Advantages and Disadvantages of Server Side and Client-Side Scripting**

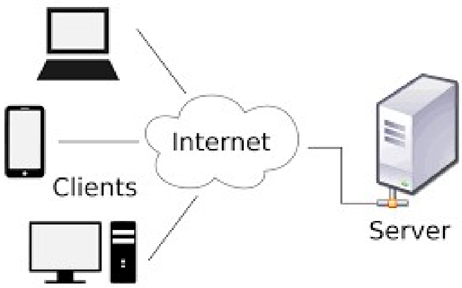
https://www.geeksforgeeks.org/difference-between-server-side-scripting-and-client-side-scripting/

- **Client:** When we talk the word **Client**, it means to talk of a person or an organization using a particular service. Similarly in the digital world a **Client** is a computer (**Host**) i.e. capable of receiving information or using a particular service from the service providers (**Servers**).
- **Servers:** Similarly, when we talk the word **Servers**, It means a person or medium that serves something. Similarly in this digital world a **Server** is a remote computer which provides information (data) or access to particular services.

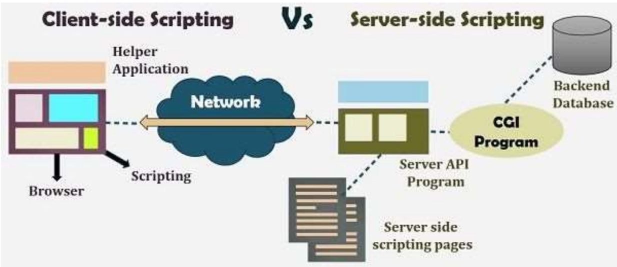
So, its basically the **Client** requesting something and the **Server** serving it as long as its present in the database.






### 6.1. Overview of Server Side and Client-Side Scripting




A software application is stored on a computer's hard drive, but a web application is stored on a server and used through a web browser.

The **client-side** (or front-end) is the user interface of the application where apps are displayed in the browser using HTML, CSS and JavaScript.

The **server-side** (or back-end) stores and processes the bulk of the data and source code using languages such as SQL, PHP and Python.

On a dynamic website there are client-side and server-side scripts. A **Script** is generally a series of programs or instructions which must be executed in other program or application. The client-side of a website refers to the web browser and the server-side is where the data and source code is stored.

The scripts can be written in two forms, at the server end/ server-side (back end) or at the client end/ Client-side (front end). The main difference between server-side scripting and client-side scripting is that the server-side scripting involves server for its processing. On the other hand, client-side scripting requires browsers to run the scripts on the client machine but does not interact with the server while processing the client-side scripts.

**Client-side scripts -**

A client-side script is a program that is processed within the client browser. These kinds of scripts are small programs which are downloaded, compiled/interpreted and run by the browser. JavaScript is an important client-side scripting language and widely used in dynamic websites. The script can be embedded within the HTML or stored in an external file.

External scripts are sent to the client from the server when they are requested. Scripts can also be executed as a result of the user doing something like pressing a page button.

Client-side scripts can often be looked at if the user chooses to view the source code of the page. JavaScript code is widely copied and recycled.

**Server-side scripts -**

A server-side script is processed on the web server when the user requests information. These kinds of scripts can run before a web page is loaded. They are needed for anything that requires dynamic data, such as storing user login details. Some common server-side languages include PHP, Python, Ruby and Java. These execute like programming languages on the server.

When a server-side script is processed, the request is sent to the server and the result is sent back to the client. This is useful for websites which store large amounts of data, such as search engines or social networks - it would be very slow for the client browser to download all the data.

### 6.2. Difference between Server Side and Client-Side Scripting

| BASIS FOR COMPARISON | SERVER-SIDE SCRIPTING | CLIENT-SIDE SCRIPTING |
|---|---|---|
| Basic | Works in the back end which could not be visible at the client end. | Works at the front end and script are visible among the users. |
| Processing | Requires server interaction. | Does not need interaction with the server. |
| Languages involved | PHP, ASP.net, Ruby on Rails, ColdFusion, Python, etcetera. | HTML, CSS, JavaScript, etc. |
| Affect | Could effectively customize the web pages and provide dynamic websites. | Can reduce the load to the server. |
| Security | Relatively secure. | Insecure |

## COMPARISON

| Client side scripting | Server side scripting |
|---|---|
| Used when the users browser already has all the code | Used to create dynamic pages |
| The Web Browser executes the client side scripting | The Web Server executes the server side scripting |
| Cannot be used to connect to the databases on the web server | Used to connect to the databases that reside on the web server |
| Can't access the file system that resides at the web server | Can access the file system residing at the web server |
| Response from a client-side script is faster as compared to a server-side script | Response from a server-side script is slower as compared to a client-side script |

### 6.3. Advantages and Disadvantages of Server Side and Client-Side Scripting

**Assignment**

https://statesmaneffiomedem.wordpress.com/advantages-and-disadvantages-of-client-side-and-server-side-scripting/

#### Client-Side Scripting

##### Advantages

- The response time is often quicker.
- Reduces the web server's workload.

##### Disadvantages

- Browser specific
- The code in your client-side scripts is completely visible to the user.

#### Server Side Scripting

##### Advantages

- Allows running programs in programming languages that aren't supported by the browser.
- Enables you to program dynamic web applications browser- independently,
- Client-side programming features such as Java applets, Dynamic Html, and ActiveX controls, all of which are browsed specifically.
- Can provide the client (browser) with data that does not reside at the client.

- Provides improved security measures, since you can write code that can never be viewed from the browser.

### Disadvantages

- Increases the workload of the server.